

Adversarial-Resilient GNSS Positioning for Denied and Degraded Environments: NavIC-SHIELD

BSERC Def-Space Summer Internship Project Report

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Abstract

GNSS spoofing poses a significant navigation-integrity risk to defence, autonomous, and border-surveillance systems because it can manipulate a receiver's position solution without producing an obvious loss of service. NavIC-SHIELD is an end-to-end research framework that simulates a NavIC constellation under terrain-aware visibility constraints, introduces controlled step, drift, and evasive spoofing attacks, extracts temporal and spatial signal features, performs satellite-level threat detection through spatio-temporal fusion, aggregates this evidence into an epoch-level navigation integrity confidence, and applies a confidence-aware Kalman fallback to protect the navigation solution. Experimental evaluation shows that the fusion detector achieves perfect recall (1.000) against the step attack and improves recall from 0.967 to 0.989 against the drift attack relative to a temporal-only baseline, while the evasive attack remains a demonstrated limitation (fusion recall 0.089). At the navigation level, Kalman-based protection reduces mean position error from 145.82 m to 1.90 m under the step attack and, most notably, from 6289.44 m to 2.94 m under the drift attack, with a smaller but still positive reduction from 29.10 m to 18.40 m under the evasive attack. An interactive Streamlit dashboard provides constellation intelligence, satellite-wise threat inspection, navigation-assurance visualizations, and detector-performance comparisons for interactive demonstration.

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1 Introduction

Global Navigation Satellite Systems (GNSS) have become a fundamental component of modern navigation, positioning, timing, autonomous systems, transportation, surveying, communication infrastructure, and defence applications. Systems operating in remote, mountainous, border, and contested environments increasingly depend on reliable positioning information to maintain situational awareness and make navigation decisions. However, the signals received by GNSS receivers are inherently weak and are transmitted through open interfaces, making them susceptible to intentional interference and manipulation. Among these threats, spoofing is particularly dangerous because it does not necessarily result in an obvious loss of navigation service. Instead, an adversary can manipulate the measurements received by a navigation receiver such that the receiver continues to produce apparently valid position estimates while gradually or abruptly being driven toward an incorrect solution.

This makes GNSS spoofing fundamentally different from a conventional signal outage. During an outage, the receiver may recognize that navigation information is unavailable and transition to an alternative positioning mechanism. During spoofing, the receiver may continue to believe that its measurements are trustworthy, so a navigation system can become unreliable without producing an immediately obvious warning. This creates a significant navigation integrity problem for defence platforms, autonomous vehicles, UAVs, border-surveillance systems, and other systems operating in environments where incorrect positioning can have serious consequences.

NavIC-SHIELD was developed as a research framework to investigate this problem through an end-to-end simulation, detection, assessment, and navigation-protection pipeline. The project focuses on the use of NavIC/GNSS observations, temporal and spatial signal behaviour, satellite-level threat assessment, spatio-temporal evidence fusion, and confidence-aware navigation protection. Rather than treating spoofing detection only as a binary classification problem, NavIC-SHIELD connects detection decisions with the actual navigation solution: identifying suspicious satellite behaviour ultimately contributes to determining whether the current GNSS position can be trusted and whether an alternative positioning strategy should be used.

2 Problem Statement, Aim, and Objectives

2.1 Problem Statement

GNSS receivers normally estimate their position using measurements obtained from multiple visible satellites. Under normal conditions, these measurements exhibit relationships governed by satellite geometry, receiver position, signal propagation, and measurement noise. Spoofing attacks attempt to disrupt these relationships by introducing manipulated measurements. If the manipulation is sufficiently coordinated or gradual, a receiver may accept the corrupted observations and calculate an incorrect but mathematically plausible position.

A major challenge is therefore to distinguish natural measurement variation and environmental degradation from deliberate or adversarial manipulation. Terrain, satellite visibility, satellite geometry, signal quality, and receiver conditions can naturally produce changes in GNSS measurements. A robust detection framework must avoid interpreting every deviation as an attack while remaining sensitive to coordinated or persistent anomalies.

NavIC-SHIELD addresses this challenge by constructing a controlled environment in which normal and adversarial GNSS behaviour can be studied systematically. The framework models a NavIC constellation, applies terrain-aware visibility constraints, generates GNSS observations, introduces controlled spoofing scenarios, extracts temporal and spatial characteristics, estimates satellite-level threat probabilities, and combines these signals to produce a navigation integrity assessment. The resulting confidence information is then used to influence the

navigation solution through a Kalman-based fallback mechanism.

2.2 Aim of the Project

The primary aim of NavIC-SHIELD is to develop and evaluate an adversarial-resilient GNSS positioning framework capable of detecting suspicious satellite behaviour and reducing the effect of compromised measurements on receiver positioning.

The project is designed around the principle that a navigation protection system should not stop at identifying an anomaly. It should also determine the potential impact of that anomaly on navigation integrity and provide a mechanism for maintaining a more stable position estimate when confidence in the GNSS solution decreases. The resulting framework therefore follows a broader **Detect** → **Assess** → **Protect** approach: the detection stage identifies anomalous signal behaviour, the assessment stage estimates the trustworthiness of the navigation measurements, and the protection stage determines whether the conventional GNSS position should be trusted or whether a Kalman-based fallback should be used.

2.3 Project Objectives

1. Develop a controlled NavIC/GNSS simulation environment capable of representing satellite visibility and receiver observations under realistic geometric and terrain constraints, providing a reproducible environment for evaluating navigation integrity without depending exclusively on live attack data.
2. Model different forms of spoofing behaviour, including controlled step, drift, and evasive attack scenarios, representing different temporal characteristics and levels of detectability.
3. Extract information from both temporal and spatial relationships between GNSS observations — temporal analysis examining how measurements evolve over successive epochs, and spatial analysis considering inconsistencies between different visible satellites.
4. Estimate spoofing probability at the individual satellite level, allowing the system to move beyond an overall attack/no-attack decision and identify which satellites are contributing suspicious measurements.
5. Combine satellite-level evidence into an epoch-level navigation integrity assessment, providing a bridge between signal-level anomaly detection and the navigation solution.
6. Evaluate whether confidence-aware navigation protection can reduce positioning errors when GNSS measurements are compromised, by comparing raw GNSS positioning against a protected solution incorporating Kalman fallback behaviour.

2.4 Key Contributions

The primary contributions of this work are summarized below.

1. **Terrain-aware NavIC/GNSS simulation.** A controlled simulation environment incorporating elevation-mask-based visibility constraints, providing a realistic and reproducible baseline for navigation-integrity experiments rather than an idealized clear-sky assumption.
2. **Controlled multi-mode spoofing framework with hidden ground truth.** Step, drift, and evasive attack models covering abrupt, gradual, and detection-evasive manipulation strategies, with attack state retained only for offline evaluation and never exposed to the detection pipeline.

3. **Satellite-level temporal-spatial fusion detection.** Rather than producing a single global attack/no-attack decision, the framework estimates spoof probability per satellite per epoch by fusing temporal and spatial evidence, enabling more granular and interpretable threat attribution.
4. **Confidence-based bridge from detection to navigation protection.** Satellite-level threat information is aggregated into an epoch-level navigation integrity confidence score, which directly governs a Kalman-based fallback mechanism – connecting signal-level detection to an actual navigation-level outcome rather than treating detection as an isolated classification task.
5. **Integrated evaluation and interactive visualization.** An interactive Streamlit dashboard exposing constellation intelligence, satellite-wise threat inspection, navigation assurance, and detector performance comparison, supporting both experimentation and interactive demonstration.

3 Proposed System Architecture

NavIC-SHIELD implements an end-to-end pipeline beginning with a simulated NavIC constellation, proceeding through observation generation, controlled spoofing injection, temporal-spatial feature extraction, spatio-temporal fusion detection, satellite-level threat assessment, navigation integrity confidence estimation, and finally confidence-aware Kalman-based navigation protection. The architecture was selected to directly connect signal-level detection with the navigation solution, rather than treating spoofing detection as an isolated classification task.

3.1 Pipeline Overview

1. **Constellation Simulation:** Provides the satellite states required for generating observations.
2. **Terrain-Aware Visibility:** Determines which satellites are actually observable given elevation and terrain constraints, so the receiver does not unrealistically assume continuous availability of every satellite.
3. **Observation Generation:** Produces baseline GNSS measurements for the receiver under normal conditions.
4. **Controlled Spoofing Injection:** Introduces step, drift, or evasive attacks into selected observations during defined attack windows, with ground-truth attack state retained for detector evaluation.
5. **Temporal and Spatial Feature Extraction:** Captures persistence and change in individual satellite behaviour over time, and relationships between simultaneously observed satellites.
6. **Spatio-Temporal Fusion Detection:** Combines temporal and spatial evidence into a per-satellite spoof probability.
7. **Satellite-Level Threat Assessment:** Assigns threat information to individual satellites rather than producing a single global attack label, exposed through the interactive dashboard.
8. **Navigation Integrity Confidence:** Aggregates satellite-level threat information into an epoch-level confidence score representing whether the current GNSS measurements can be trusted.

9. **Kalman-Based Navigation Protection:** When confidence falls below a configured threshold, a Kalman fallback mechanism reduces dependence on the potentially corrupted GNSS solution to maintain a more stable position estimate.

This architecture was chosen over a simpler binary detect/no-detect design for three reasons. First, satellite-level attribution allows the system to identify which satellites are contributing suspicious measurements, providing finer-grained evidence for the navigation-integrity decision. Second, combining temporal and spatial evidence allows the detector to catch attacks that are not clearly visible in either dimension alone. Third, connecting detection output to an explicit confidence score allows the navigation layer to use continuous confidence-dependent measurement weighting together with a threshold-based protection decision.

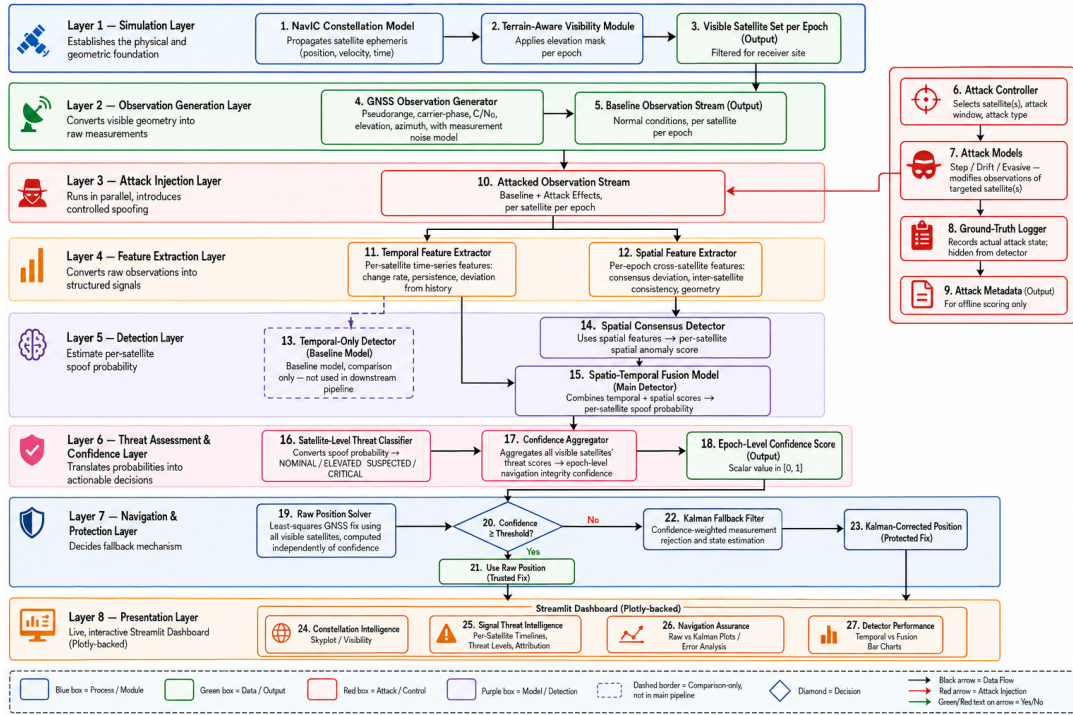


Figure 1: End-to-end NavIC-SHIELD pipeline: simulation, spoofing, detection, and navigation protection.

4 Terrain-Aware GNSS Simulation

A key component of the project is the simulation of satellite visibility under terrain constraints. In an unrestricted simulation, a receiver may be able to observe satellites that would not actually be usable because of terrain obstruction or insufficient elevation, producing an unrealistically favourable observation set.

NavIC-SHIELD therefore incorporates terrain-aware visibility into the observation generation process. Satellite elevation and terrain constraints are considered before observations are made available to the receiver, allowing the resulting dataset to better represent the relationship between the physical environment and navigation availability.

4.1 Relevance to the Defence and Border Context

This aspect is particularly relevant to the intended defence and border-environment context of the project. Mountainous and terrain-constrained environments can introduce natural degradation in satellite visibility and geometry. A spoofing detection system must therefore operate in

the presence of legitimate environmental variation rather than assuming that every abnormal measurement is caused by an adversary.

4.2 Role in the Experimental Design

The simulation consequently provides two important capabilities:

- It establishes a controlled baseline for normal GNSS behaviour under realistic visibility constraints.
- It provides a controlled environment in which adversarial manipulation can be introduced and evaluated against that baseline, with ground-truth attack state retained for detector evaluation.

Without terrain-aware visibility, a detector could be inadvertently evaluated on a constellation geometry that never occurs in practice, which would overstate detection performance relative to a real deployment.

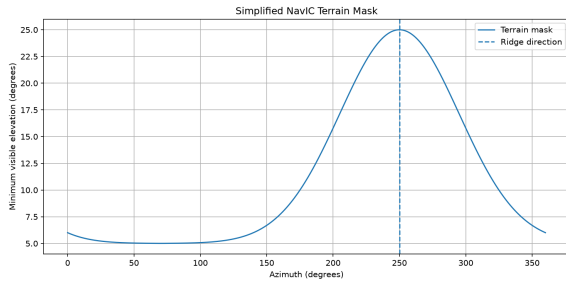


Figure 2: Simplified NavIC terrain mask used for visibility gating, showing minimum visible elevation as a function of azimuth. The peak (approximately 25°) corresponds to the ridge direction, raising the minimum usable elevation for satellites observed through that bearing.

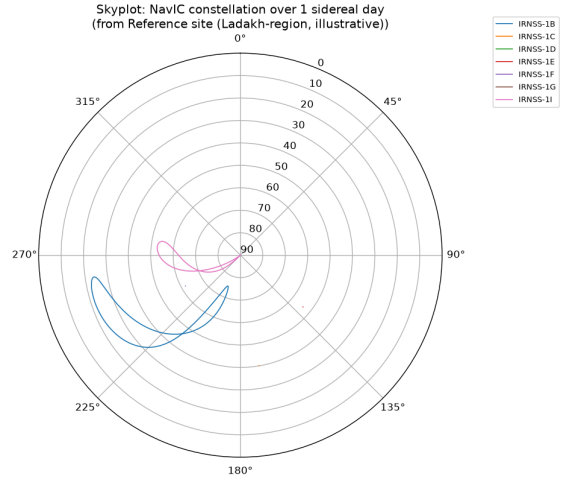


Figure 3: Skyplot of the simulated NavIC constellation over one sidereal day from the reference site, showing satellite tracks in azimuth/elevation. Only a subset of satellites (IRNSS-1B, IRNSS-1I) achieve sustained visibility above the terrain mask at this site, illustrating the effect of terrain-aware gating on effective constellation geometry.

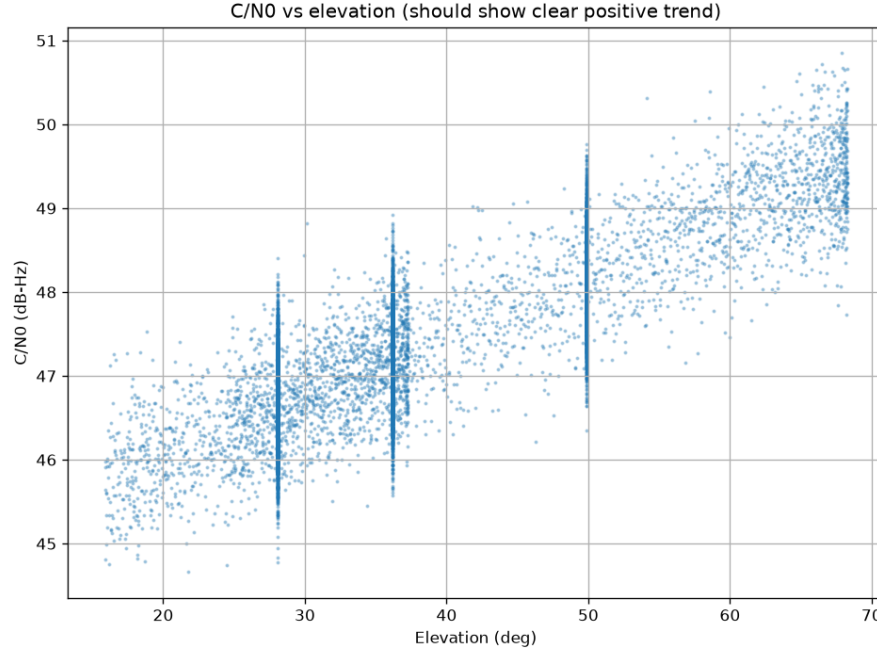


Figure 4: Carrier-to-noise density (C/N_0) versus elevation angle across simulated observations, showing the expected positive trend of increasing signal strength with elevation. This relationship is consistent with expected GNSS signal-quality behaviour and provides a sanity check for the observation-generation model prior to attack injection.

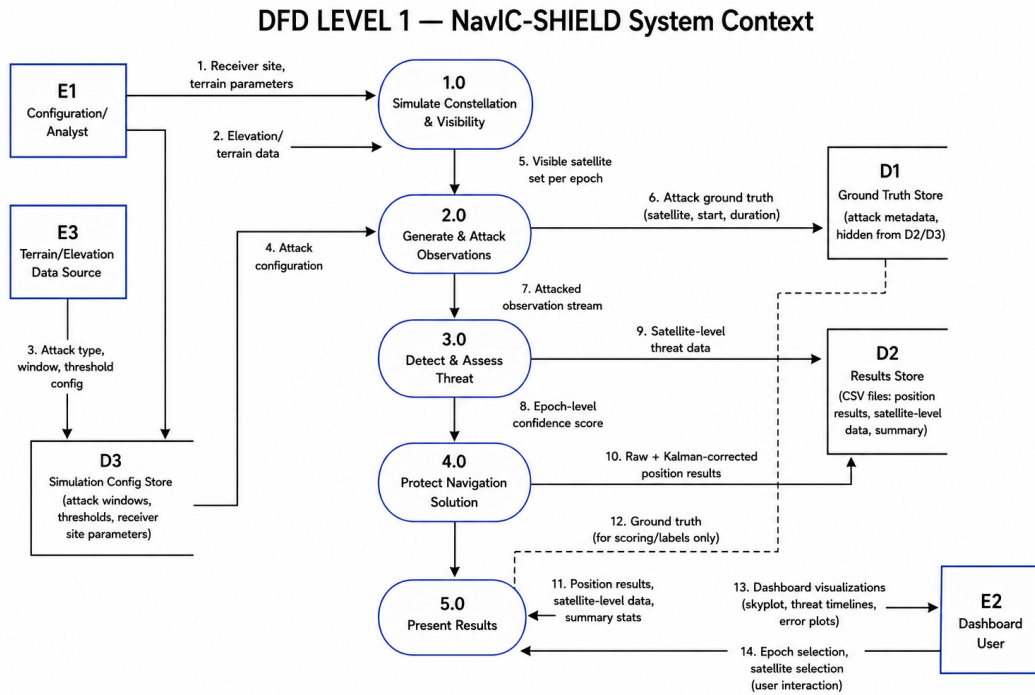


Figure 5: Data Flow Diagram (Level 1): NavIC-SHIELD system context and major data flows.

5 Spoofing Attack Models

NavIC-SHIELD evaluates three primary controlled spoofing scenarios: step, drift, and evasive attacks. Each represents a distinct temporal manipulation strategy and a different level of detectability, allowing the detection and navigation-protection pipeline to be evaluated across a realistic range of adversarial behaviour.

5.1 Step Attack

The step attack represents an abrupt measurement manipulation. The attack produces a sudden change in the affected GNSS measurement and therefore represents a relatively obvious form of spoofing. This scenario is useful for evaluating whether the detection pipeline can identify sudden anomalies quickly, and serves as a baseline case in which detection is expected to be straightforward.

5.2 Drift Attack

The drift attack represents a gradually increasing manipulation. Instead of producing a single abrupt change, the measurement is progressively altered over time. This creates a more challenging detection problem because individual observations may remain relatively close to their previous values while the cumulative effect can become significant. The drift scenario is particularly important for navigation integrity because gradual manipulation can cause the position solution to move substantially while remaining superficially plausible at any single epoch.

5.3 Evasive Attack

The evasive attack represents a more difficult detection scenario in which the manipulation is designed to remain less distinguishable from normal variation. The attack evolves in a way that makes immediate detection more difficult. This scenario is used to evaluate the limitations of the current detector and to identify areas where more advanced temporal modelling may be required.

5.4 Attack Injection Summary

Table 1: Controlled spoofing attack scenarios modeled in NavIC-SHIELD.

Attack Type	Temporal Character	Detectability
Step	Abrupt, single discontinuity	High
Drift	Gradual, monotonic ramp	Moderate
Evasive	Gradual, designed to mimic noise	Low

Modeling all three attack types within the same simulation and detection framework allows a direct, controlled comparison of detector behaviour across abrupt, gradual, and adversarially-evasive manipulation, rather than evaluating the system against only a single, most-favourable attack type.

6 Temporal and Spatial Detection Framework

The detection framework uses complementary temporal and spatial information, combined through spatio-temporal fusion, to estimate satellite-level spoofing probability.

6.1 Temporal Analysis

Temporal analysis examines the behaviour of observations across successive epochs. This enables the system to identify persistent deviations, abrupt changes, gradual trends, and other temporal characteristics associated with spoofing. A temporal-only detector was implemented as a baseline against which the benefit of adding spatial evidence could be measured.

All temporal features are computed on the pseudorange *residual* rather than the raw pseudorange, since raw pseudorange for a GSO satellite changes by thousands of metres per sample from orbital motion alone, which would swamp any injected attack offset:

$$r_{s,t} = \rho_{s,t} - \hat{\rho}_{s,t} \quad (1)$$

where $\rho_{s,t}$ is the measured pseudorange and $\hat{\rho}_{s,t}$ is the true geometric range for satellite s at epoch t . From this residual, a rolling window of $W = 8$ visible samples (≈ 4 minutes at a 30 s sampling interval) yields the first difference, rolling mean, rolling standard deviation, and a normalized deviation ("z-score"):

$$\Delta r_{s,t} = r_{s,t} - r_{s,t-1} \quad (2)$$

$$\mu_{s,t} = \text{mean}(r_{s,t-W+1}, \dots, r_{s,t}) \quad (3)$$

$$\sigma_{s,t} = \text{std}(r_{s,t-W+1}, \dots, r_{s,t}) \quad (4)$$

$$z_{s,t} = \begin{cases} \frac{r_{s,t} - \mu_{s,t}}{\sigma_{s,t}} & \sigma_{s,t} > 10^{-6} \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

together with the equivalent first-difference and rolling-slope features computed on carrier-to-noise density (C/N_0), giving the temporal feature vector $\mathbf{S}_{s,t}^{\text{temp}}$ for satellite s at epoch t .

6.2 Spatial Analysis

Spatial analysis considers the relationship between multiple satellites observed during the same navigation epoch. Because a receiver's position solution is determined using multiple satellite measurements simultaneously, abnormal behaviour in one or more satellites can create inconsistencies relative to the rest of the constellation that are not visible from any single satellite's time history alone.

At each epoch t , the spatial deviation of satellite s is measured against the *median* residual across all other satellites visible at that epoch:

$$\tilde{r}_t = \text{median}_{j \in V_t}(r_{j,t}), \quad d_{s,t} = |r_{s,t} - \tilde{r}_t| \quad (6)$$

where V_t is the set of satellites visible at epoch t . The median is used rather than the mean specifically so that a single spoofed satellite does not pull the consensus reference toward itself; this remains valid provided spoofed satellites are a minority of V_t at any given epoch, which is enforced by construction in the controlled attack scenarios (Section 5). Epochs with fewer than two visible satellites yield no spatial feature, since no consensus can be formed.

6.3 Motivation for Fusion

The use of both dimensions is important because different attacks may be detectable through different characteristics:

- A signal may not appear highly abnormal when considered independently over time, but may become suspicious when compared against other satellites observed at the same epoch.

- Conversely, a persistent temporal change may be detectable even when spatial relationships remain relatively consistent.

NavIC-SHIELD therefore combines these sources of evidence through spatio-temporal fusion. The objective is not simply to increase the number of detected anomalies, but to provide a more informative and reliable estimate of satellite and navigation integrity than either dimension could provide alone.

6.4 Fusion Model

The temporal and spatial feature sets are used to train a fusion detection model that produces a per-satellite spoof probability at each epoch. This probability is the quantity subsequently used for satellite-level threat assessment (Section 7) and for aggregation into the epoch-level navigation integrity confidence (Section 7.2).

The detection pipeline is a two-stage stacked model. First, a Random Forest classifier ϕ is trained on the temporal feature vector alone (this is also the temporal-only baseline evaluated in Section 8):

$$p_{s,t}^{\text{temp}} = \phi(\mathbf{S}_{s,t}^{\text{temp}}) \in [0, 1] \quad (7)$$

The fusion model is a logistic regression f_w trained on the concatenation of this temporal probability and the spatial deviation:

$$P_{s,t}^{\text{spoof}} = f_w([p_{s,t}^{\text{temp}}, d_{s,t}]) = \sigma(w_0 + w_1 p_{s,t}^{\text{temp}} + w_2 d_{s,t}) \quad (8)$$

where $\sigma(\cdot)$ is the logistic sigmoid and w_0, w_1, w_2 are the fitted logistic regression coefficients. This stacked design was chosen because the temporal Random Forest and the spatial consensus feature are each already independently informative; the fusion stage's role is to combine two already-strong scores rather than to learn a full representation from raw features.

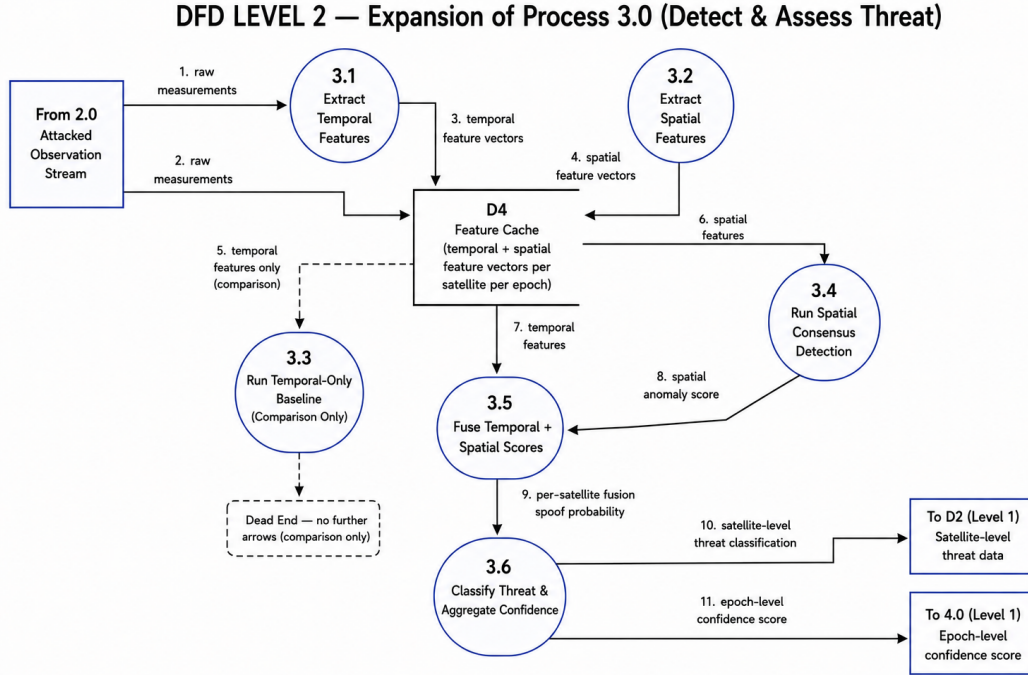


Figure 6: Data Flow Diagram (Level 2): Temporal and spatial feature extraction and spatio-temporal fusion.

7 Satellite-Level Threat Intelligence and Navigation Integrity Confidence

7.1 Satellite-Level Threat Intelligence

One of the important features introduced in the project is satellite-wise threat intelligence. Instead of presenting only an overall navigation status, the dashboard allows the user to select an individual satellite and inspect its available threat information, providing a more interpretable view of the detection system.

Satellite-level analysis is important because GNSS navigation is inherently multi-satellite. If only a subset of satellites is compromised, treating the entire constellation as equally untrusted can unnecessarily reduce navigation availability. Conversely, if several satellites exhibit coordinated suspicious behaviour, the navigation solution may become significantly less reliable. The satellite-wise view therefore provides an intermediate layer between raw signal measurements and the overall navigation decision, helping identify which observations are contributing to a detected threat.

Each satellite’s fusion spoof probability $P_{s,t}^{\text{spoof}}$ (Section 6.4) is mapped to a discrete threat level using fixed threshold bands:

$$T_{s,t} = \begin{cases} \text{CRITICAL} & P_{s,t}^{\text{spoof}} \geq 0.80 \\ \text{SUSPECTED} & 0.50 \leq P_{s,t}^{\text{spoof}} < 0.80 \\ \text{ELEVATED} & 0.25 \leq P_{s,t}^{\text{spoof}} < 0.50 \\ \text{NOMINAL} & P_{s,t}^{\text{spoof}} < 0.25 \end{cases} \quad (9)$$

7.2 From Detection to Navigation Integrity

The project extends the detection pipeline by converting satellite-level threat information into navigation integrity confidence. This is a key distinction between NavIC-SHIELD and a conventional anomaly-detection demonstration: a detector may identify suspicious measurements, but the ultimate question for a navigation system is whether the current position estimate should be trusted.

NavIC-SHIELD therefore uses the detected threat information to assess the reliability of the current GNSS navigation state. The threat levels of all satellites visible at epoch t are aggregated into a single epoch-level navigation integrity confidence score $C_t \in [0, 1]$. The resulting confidence mechanism provides a connection between signal intelligence and navigation protection:

- **High confidence** ($C_t \geq C_{\text{threshold}}$) permits continued reliance on GNSS positioning.
- **Low confidence** ($C_t < C_{\text{threshold}}$) indicates that the navigation solution may be affected by compromised measurements, and triggers the Kalman-based protection mechanism described in Section 9. The configured reject threshold is $C_{\text{threshold}} = 0.15$.

This design allows the detection system to become part of the navigation decision-making process rather than functioning as an isolated classification module. The confidence score computed at this stage is the same quantity later compared against the reject threshold in the navigation performance evaluation (Section 9.2).

8 Experimental Evaluation: Detection Performance

This section documents the methodology, assumptions, and results of the detection-performance evaluation.

8.1 Methodology

The framework was evaluated using controlled attack scenarios representing step, drift, and evasive spoofing behaviour, as defined in Section 5. For each scenario, a temporal-only detector (Section 6.1) was compared against the combined temporal–spatial fusion detector (Section 6.4) using the same underlying observation set, ensuring that any performance difference reflects the value of the spatial evidence channel rather than a difference in input data.

8.2 Evaluation Conditions and Assumptions

- **Ground truth:** Attack state (which satellite, start time, and duration) is retained internally during simulation and used only for scoring, not exposed to the detector.
- **Metrics:** Recall and F1 score were used as the primary detection metrics, since the operational concern is correctly identifying spoofed observations (recall) while maintaining a reasonable precision/recall balance (F1).
- **Per-attack evaluation window:** Each attack type was scored over its own observation window (step and drift over a 240-epoch window; evasive over a 90-epoch window), reflecting the differing configured duration of each controlled attack scenario.

8.3 Results and Discussion

Table 2: Detection performance: temporal-only baseline vs. spatio-temporal fusion.

Attack Type	Temporal Recall	Temporal F1	Fusion Recall	Fusion F1
Step	1.000	1.000	1.000	1.000
Drift	0.967	0.983	0.989	0.994
Evasive	0.111	0.200	0.089	0.163

For the step attack, both the temporal-only detector and the fusion detector achieved a recall and F1 score of 1.000, indicating that the abrupt attack was consistently and reliably identifiable regardless of whether spatial evidence was included.

For the drift attack, the temporal-only detector achieved a recall of 0.967 and F1 score of 0.983. With spatio-temporal fusion, recall increased to 0.989 and F1 increased to 0.994. This demonstrates that spatial evidence provided a measurable improvement in identifying gradually evolving spoofing behaviour, beyond what temporal information alone could capture.

The evasive attack remained the most difficult scenario. Temporal-only detection achieved a recall of 0.111 and F1 score of 0.200, while the fusion approach achieved a recall of 0.089 and F1 score of 0.163. Row-level recall alone somewhat understates the practical picture: the evasive attack is specifically designed to remain below per-row detection thresholds during an extended ramp, so detection *latency* – how long the system takes to raise sustained suspicion – is a more operationally relevant metric than row-level recall in isolation. Even accounting for this, the results indicate that the current framework does not yet reliably identify this type of subtle adversarial behaviour at the row level, and this limitation is treated explicitly rather than concealed (see Section 12).

Spatial detects the evasive attack 37.0 minutes earlier than temporal-only

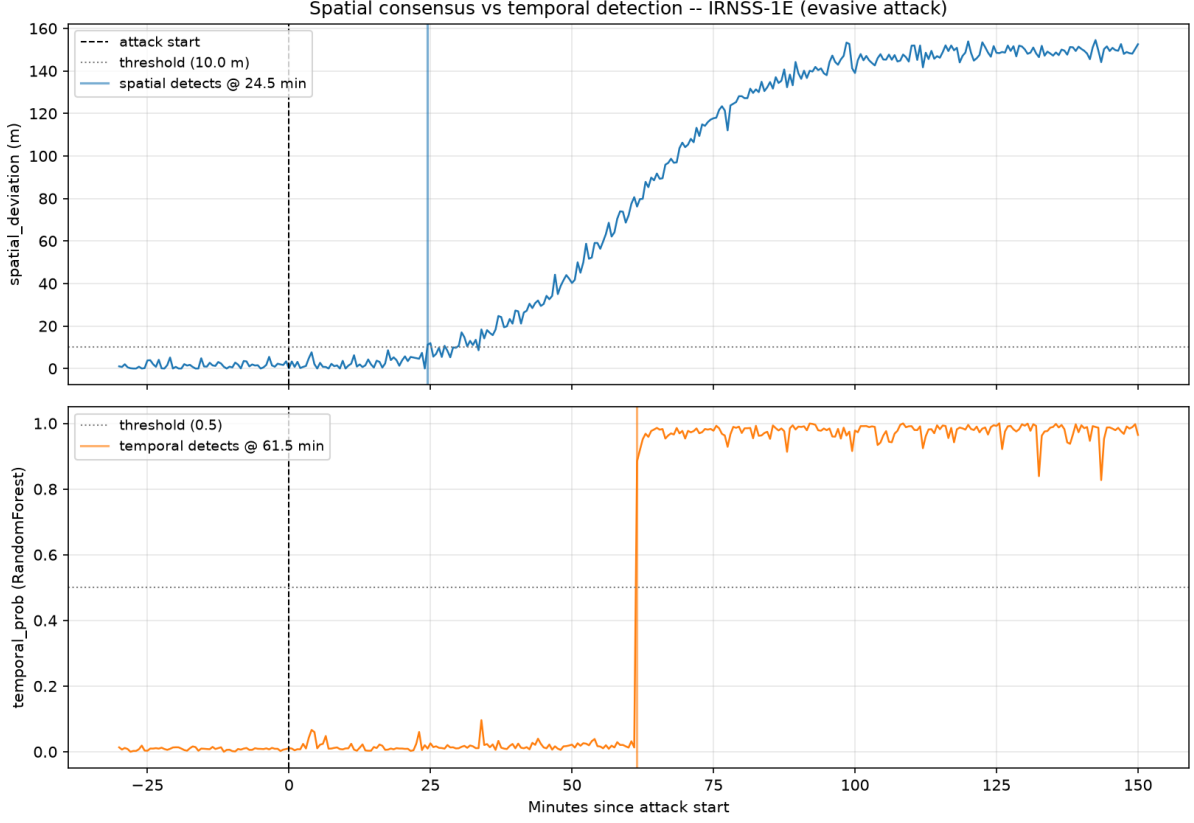


Figure 7: Detection latency comparison for the evasive attack on satellite IRNSS-1E. The spatial-consensus signal crosses its detection threshold at 24.5 minutes after attack start, while the temporal-only detector does not cross its threshold until 61.5 minutes – a 37-minute latency advantage for spatial evidence. This is a separate per-satellite latency diagnostic rather than the row-level recall metric in Table 2, and directly supports the claim that detection latency, not row-level recall alone, is the more operationally meaningful metric for this attack type.

8.4 Limitations of the Detection Evaluation

The detection evaluation is performed entirely within the controlled simulation environment described in Sections 3–5. Real GNSS signals contain additional sources of variation — multipath, atmospheric effects, receiver clock noise, and hardware-specific artifacts — that are not fully represented here. The reported recall and F1 values should therefore be interpreted as conceptual-level performance indicators rather than certified operational metrics, consistent with the research-prototype status of the project (Section 12).

9 Navigation Performance Evaluation

9.1 Kalman-Based Navigation Protection

When GNSS observations become unreliable, NavIC-SHIELD uses a Kalman-based fallback mechanism to provide a more stable navigation estimate. The purpose of this component is not to claim that Kalman filtering eliminates spoofing; rather, it provides a mechanism for maintaining a smoother navigation state when confidence in the direct GNSS solution becomes low, as determined by the navigation integrity confidence described in Section 7.2.

The filter maintains a position-only state, deliberately excluding velocity, since the receiver site is known to be stationary:

$$\mathbf{x}_k = \begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix} \in \mathbb{R}^3 \quad (\text{ECEF coordinates}) \quad (10)$$

The state transition is the identity ($F = I$), so prediction only grows the state covariance through process noise:

$$\mathbf{x}_{k|k-1} = \mathbf{x}_{k-1|k-1}, \quad P_{k|k-1} = P_{k-1|k-1} + Q_k, \quad Q_k = \sigma_p^2 \Delta t I_3 \quad (11)$$

where σ_p is the process-noise standard deviation and Δt is the time since the previous update. An earlier constant-velocity formulation was found to extrapolate spurious noise-driven velocity indefinitely once measurement updates were gated off during a sustained attack, producing unbounded runaway; the constant-position model avoids this failure mode.

The filter applies two layers of protection against a compromised measurement. First, the measurement noise is scaled continuously with the navigation confidence C_t :

$$R_k = (\sigma_{\text{base}} \cdot [m_{\text{max}} + C_t(m_{\text{min}} - m_{\text{max}})])^2 I_3 \quad (12)$$

where σ_{base} is the base measurement-noise standard deviation and $m_{\text{min}}, m_{\text{max}}$ bound the confidence-dependent noise multiplier, so that low confidence inflates R_k and proportionally reduces the filter's trust in the incoming measurement. Second, a hard reject gate skips the measurement update entirely whenever $C_t < C_{\text{threshold}} = 0.15$, since soft down-weighting alone still permits a small, nonzero Kalman gain to accumulate toward a persistently biased measurement over many repeated updates. When a measurement is accepted, the standard Kalman update is applied:

$$\mathbf{y}_k = \mathbf{z}_k - \mathbf{x}_{k|k-1}, \quad S_k = P_{k|k-1} + R_k, \quad K_k = P_{k|k-1} S_k^{-1} \quad (13)$$

$$\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + K_k \mathbf{y}_k, \quad P_{k|k} = (I - K_k) P_{k|k-1} \quad (14)$$

where $\mathbf{z}_k$ is the raw solved position from the least-squares position solver.

The protected solution can therefore be compared directly against the raw GNSS solution. This comparison provides a quantitative measure of whether the threat assessment and fallback mechanism actually reduce the impact of spoofed measurements on the final position estimate. This is particularly important for the drift attack, where a gradual spoofing attack can cause a large navigation error even when individual measurements do not appear catastrophically abnormal at any single epoch.

9.2 Evaluation Methodology

For each attack scenario, mean position error was computed for both the raw (uncorrected) GNSS position and the Kalman-corrected position, over the same attack window used for the corresponding detection evaluation in Section 8 (240 epochs for step and drift; 90 epochs for evasive, reflecting that attack's shorter configured ramp duration).

9.3 Results and Discussion

For the step attack, the raw navigation solution produced a mean position error of approximately 145.82 m over the 240-epoch attack window. With the protection mechanism, the mean error was reduced to approximately 1.90 m.

Table 3: Navigation performance: raw vs. Kalman-corrected mean position error.

Attack Type	Epochs	Raw Error (m)	Kalman Error (m)	Reduction
Step	240	145.82	1.90	98.7%
Drift	240	6289.44	2.94	99.95%
Evasive	90	29.10	18.40	36.8%

The drift attack produced the most significant degradation in the raw navigation solution over its 240-epoch window. Its mean position error reached approximately 6289.44 m, demonstrating how gradual manipulation can produce severe navigation corruption despite each individual measurement appearing only mildly abnormal. After applying the confidence-aware Kalman protection mechanism, the mean position error was reduced to approximately 2.94 m — a reduction of over 99.9%. This very large reduction demonstrates the mechanism’s effectiveness within the controlled experimental environment described in Sections 3–5; it should not be interpreted as equivalent to real-world protection performance, particularly given the simulation-to-reality gap discussed in Section 12.3.

For the evasive attack, evaluated over its 90-epoch window, the raw mean position error was approximately 29.10 m, while the protected solution reduced this to approximately 18.40 m. Although the improvement was smaller in absolute and relative terms than for the other attack scenarios, the result demonstrates that the protection mechanism can still meaningfully reduce navigation error even under a detection-evasive condition, despite the fusion detector’s low row-level recall for this attack (Section 8.3).

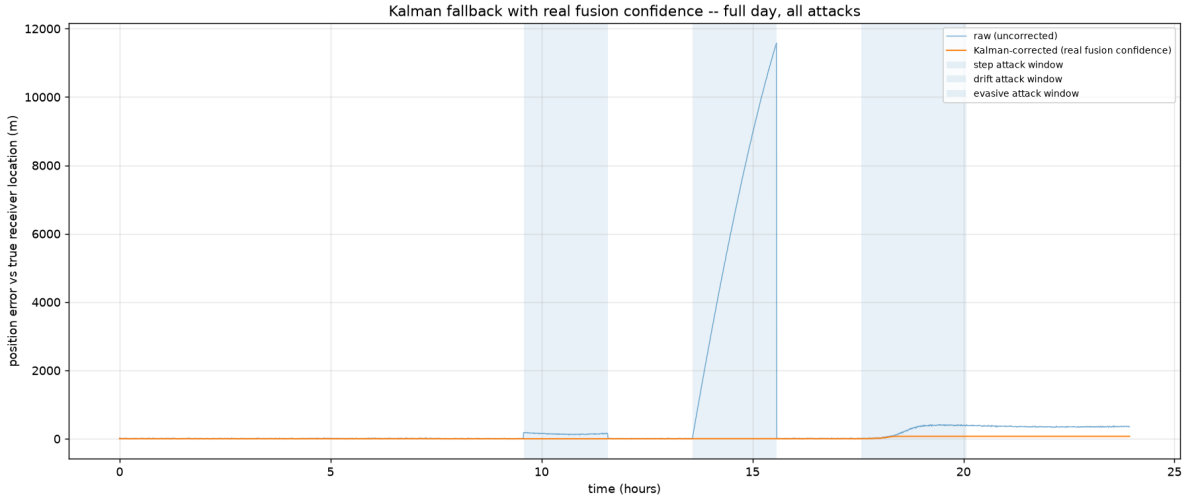


Figure 8: Full-day raw vs. Kalman-corrected position error with all three attack windows shaded. The drift attack window produces the largest raw excursion (peaking near 11,600 m), consistent with Table 3, while the Kalman-corrected trace remains near zero throughout all three attack windows. Note the evasive-attack window shown here extends beyond the 90-epoch (45-minute) segment used for the scored evaluation in Table 3; raw error continues to grow past that scoring window as the attack ramp continues.

9.4 Interpretation

These results demonstrate the importance of connecting spoofing detection with navigation protection rather than evaluating detection performance in isolation. A detector’s recall and F1 score are useful diagnostic metrics, but the ultimate operational concern is the effect of

compromised measurements on the navigation solution actually delivered to the user. Notably, the evasive attack shows that navigation-level protection can provide value even in cases where row-level detection is weak, since the confidence-aggregation and Kalman correction stages (Sections 7–9) operate on accumulated evidence rather than a single-epoch decision.

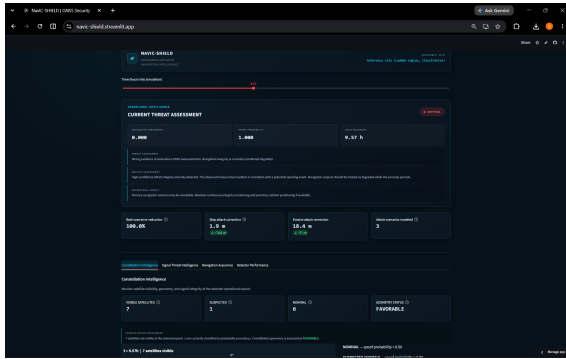
10 Interactive Dashboard

NavIC-SHIELD includes an interactive Streamlit dashboard designed to make the experimental results accessible without requiring users to inspect raw datasets or execute individual analysis scripts.

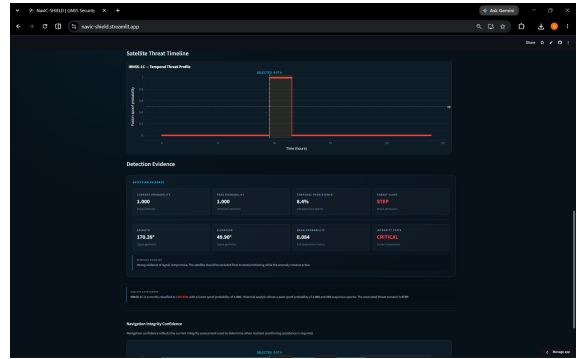
10.1 Dashboard Components

The dashboard is organized into four primary views:

- **Constellation Intelligence:** An interactive skyplot showing visible satellites at a selected epoch, with markers sized and coloured by spoof probability, alongside a satellite-selection control for inspecting individual satellite geometry and status.
- **Signal Threat Intelligence:** Per-satellite detection state, historical detection profile (peak and mean spoof probability, suspicious-epoch count), attack attribution, and a temporal threat-probability timeline with the relevant attack window highlighted.
- **Navigation Assurance:** Full-day position scatter and position-error-over-time plots comparing the raw and Kalman-corrected navigation solutions, together with the per-attack summary table shown in Section 9.
- **Detector Performance:** A grouped bar-chart comparison of temporal-only versus fusion-detector recall across attack types, corresponding to the results in Section 8.

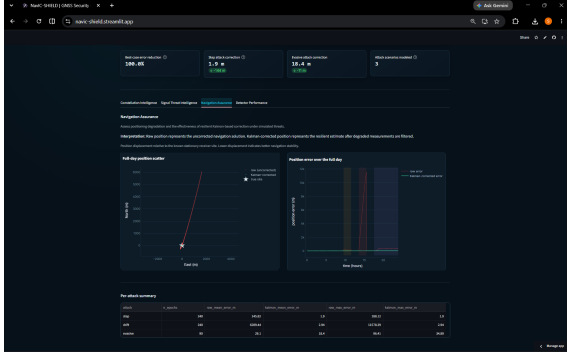


(a) Overall deployed dashboard

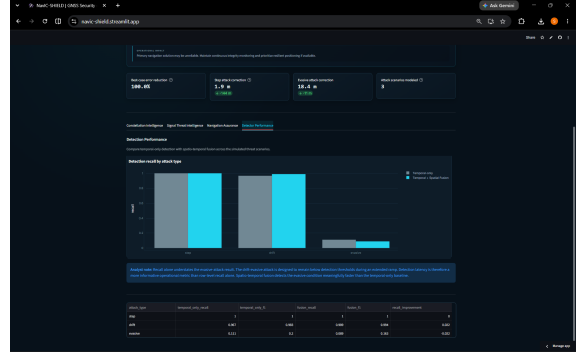


(b) Signal Threat Intelligence

Figure 9: Deployed dashboard views for overall system monitoring and satellite-level threat assessment.



(a) Navigation Assurance



(b) Detector Performance

Figure 10: Deployed dashboard views for navigation protection and detector-performance evaluation.

10.2 Purpose

The dashboard provides an interpretable interface over the underlying detection and navigation pipeline. Rather than presenting only aggregate metrics, it allows a user to inspect individual satellite behaviour and examine how satellite-level evidence contributes to the overall navigation integrity assessment described in Section 7.2. An operational threat-assessment panel further summarizes the current navigation confidence and spoof probability into a CRITICAL/HIGH/ELEVATED/NOMINAL status indicator with accompanying analyst-style commentary, making the system’s reasoning easier to interpret during an interactive demonstration.

The dashboard is implemented in Python using Streamlit for the application framework and Plotly for interactive visualization. It has been deployed using Streamlit to support remote demonstration of the NavIC-SHIELD system.

11 Technology, Implementation, and Applications

11.1 Technology Stack

The project is implemented primarily in Python. Numerical and data-processing components use NumPy and Pandas. Machine-learning functionality uses scikit-learn, with a Random Forest classifier for temporal detection and Logistic Regression for spatio-temporal fusion. Visualization and dashboard functionality use Plotly and Streamlit.

11.2 Software Architecture

The software architecture is organized into separate modules for simulation, spoofing, feature extraction, detection, navigation, and dashboard presentation, corresponding to the pipeline stages described in Section 3.1. This modular structure allows individual components to be evaluated and extended independently – for example, the spoofing module can be extended with additional attack types (Section 5) without requiring changes to the detection or navigation modules.

The project is maintained using Git and GitHub, with the source code, documentation, selected experimental outputs, and deployment configuration maintained in the project repository.

11.3 Applications and Relevance

The primary motivation for NavIC-SHIELD is the reliability of satellite navigation in environments where positioning integrity is important. Potential applications include:

- Defence navigation and border surveillance systems operating in contested or terrain-constrained environments.
- UAV and autonomous platform navigation, where an incorrect position estimate can have safety-critical consequences.
- Remote-area navigation, where alternative positioning infrastructure may be unavailable.
- GNSS integrity monitoring more broadly, as a research platform for evaluating new spoofing detection and navigation-protection algorithms under controlled, repeatable conditions.

The project is particularly relevant to situations where a navigation receiver cannot simply assume that all available satellite measurements are trustworthy. In such environments, satellite-level threat assessment and confidence-aware navigation can provide additional information for deciding when a conventional GNSS solution should be trusted, and when a resilient fallback should be engaged instead.

12 Limitations

The current implementation should be considered a research prototype rather than a certified operational navigation-protection system. The following limitations are treated as research directions rather than hidden weaknesses of the project.

12.1 Simulation-Only Evaluation

The principal evaluation is performed using a controlled simulation and controlled spoofing scenarios (Sections 3–5). While this provides reproducibility and allows systematic experimentation, real-world GNSS signals contain additional sources of variation – multipath, atmospheric effects, receiver clock noise, and hardware-specific artifacts – that may not be completely represented by the current simulation. Results reported in Sections 8–9 should therefore be interpreted as conceptual-level performance indicators rather than certified operational metrics.

12.2 Evasive-Attack Detectability

The evasive attack results (Section 8.3) demonstrate a current limitation of the detection framework. The low row-level recall obtained for this scenario (0.089 with fusion) indicates that subtle and gradually evolving adversarial behaviour remains difficult to distinguish reliably from normal measurement variation using the current temporal and spatial feature set. While the navigation-level protection mechanism still provides a positive error reduction under this condition (Section 9.3), this is a partial mitigation rather than a solution to the underlying detection gap.

12.3 Simulation-to-Reality Gap Analysis

An important limitation of the current evaluation is that all detection and navigation-protection results reported in Sections 8–9 are obtained within the controlled simulation environment described in Sections 3–5. As an initial step toward closing this gap, real GPS L1 observation

logs were used to characterize natural (non-adversarial) measurement behaviour, providing a baseline against which the simulated environment can be compared.

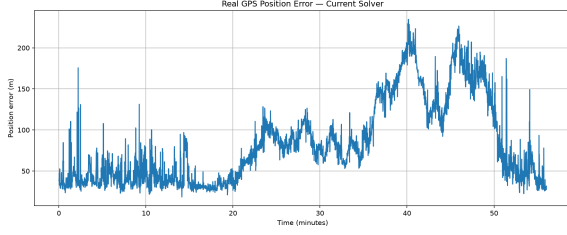


Figure 11: Real GPS position error over approximately 55 minutes using the current solver, with no injected attack. Natural error fluctuates between approximately 25 m and 230 m, substantially exceeding the simulated no-attack baseline.

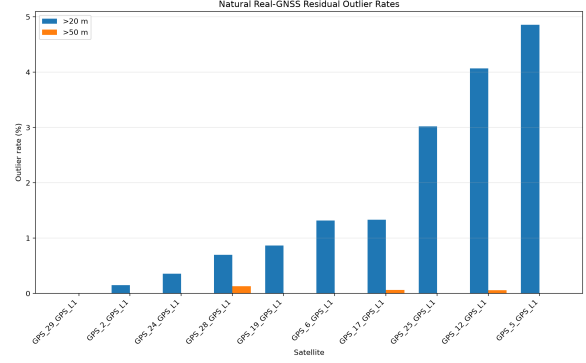


Figure 12: Natural residual outlier rates across real GPS satellites. Several satellites exceed the 20 m residual threshold on more than 3% of epochs under normal (non-spoofed) conditions.

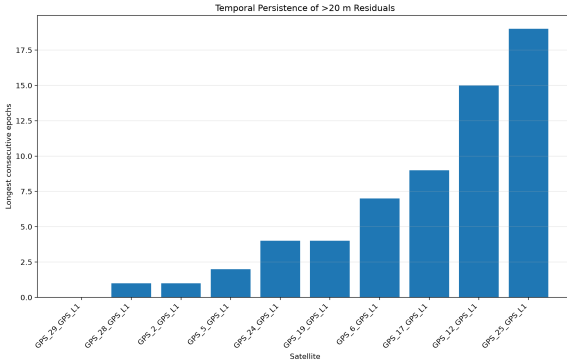


Figure 13: Longest consecutive run of >20 m residuals per satellite in real GPS data. Naturally occurring residuals can persist for up to 19 consecutive epochs on some satellites.

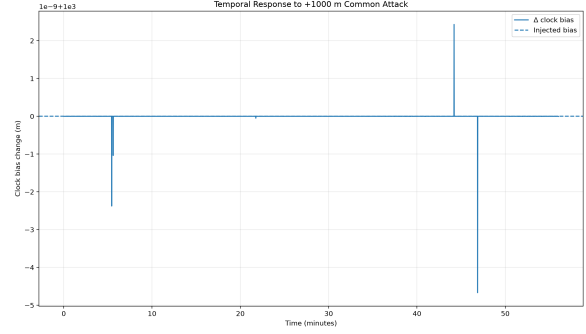


Figure 14: Preliminary real-data experiment showing receiver clock-bias response to a synthetically injected +1000 m common-mode attack, replayed against real observation timing. The response is dominated by sharp, sparse correction spikes rather than a smooth ramp.

Table 4 summarizes the principal differences observed between the simulated environment and real GPS behaviour.

Table 4: Summary of the observed simulation-to-reality gap.

Property	Simulation	Real GPS
Measurement noise	Controlled, parametric noise model	Natural, receiver- and environment-dependent
Multipath	Simplified / largely absent	Present, contributing to outliers and persistence
Clock effects	Controlled clock-bias model	Observed sharp, sparse correction spikes under injected attack
Residual magnitude	Lower, bounded by noise model	Higher; natural position error observed between 25–230 m
Residual persistence	Controlled	Naturally persists for up to 19 consecutive epochs on some satellites
Attack response domain	Primarily position-domain	Clock-domain response also observed under common-mode attack

These findings indicate a measurable simulation-to-reality gap. In particular, the natural outlier and persistence rates shown in Figs. 12 and 13 suggest that a detector calibrated purely on simulated noise statistics would likely require retuning before deployment against real receiver data, since naturally occurring residual persistence of this magnitude risks being misclassified as drift or evasive spoofing (Section 7) if the detection thresholds are not recalibrated accordingly. Similarly, the clock-domain response observed in Fig. 14 suggests that real-world attack manifestations may not be fully captured by the position-domain attack models used in the current simulation (Section 5).

This analysis is exploratory rather than a full validation, and full closure of the simulation-to-reality gap – including retraining detection models on real observation statistics and extending the attack models to cover clock-domain manifestations – remains future work (Section 14).

12.4 Objective Achievement Summary

Table 5 maps each project objective defined in Section 2.3 to its corresponding implementation, supporting evidence within this report, and completion status.

Table 5: Objective achievement matrix, mapping stated project objectives to implementation and supporting evidence.

Objective	Implementation	Evidence	Status
Terrain-aware GNSS simulation environment	NavIC constellation model with elevation-mask-based visibility filtering	Figs. 2, 3, 4	Achieved
Controlled spoofing attack modeling (step, drift, evasive)	Three distinct attack injection models with hidden ground truth	Table 1	Achieved
Temporal and spatial feature extraction	Per-satellite temporal feature extractor and per-epoch spatial consensus extractor	Section 6	Achieved
Satellite-level spoofing probability estimation	Spatio-temporal fusion model producing per-satellite spoof probability	Table 2, dashboard Signal Threat Intelligence tab	Achieved
Epoch-level navigation integrity confidence	Satellite-level threat classification aggregated into a single epoch confidence score	Section 7.2, architecture diagram (Layer 6)	Achieved
Confidence-aware navigation protection evaluation	Kalman-based fallback mechanism gated by navigation confidence threshold	Table 3, Fig. 8	Achieved

All six stated objectives were implemented and evaluated within the defined project scope. It should be noted, however, that "achieved" here refers to successful implementation and controlled-environment evaluation rather than certified real-world performance – a distinction elaborated further in Section 12 and addressed through the future-work directions in Section 14.

12.5 Scope of the Navigation Model

The Kalman-based protection mechanism (Section 9.1) is designed to maintain a smoother navigation state under low-confidence conditions; it does not itself constitute a proof that spoofing has been eliminated, only that its immediate positional impact has been reduced. Extending this to a more rigorous integrity-monitoring guarantee is left as future work (Section 14).

13 Conclusions

NavIC-SHIELD demonstrates an end-to-end approach to adversarial GNSS resilience in which spoofing detection is directly connected to navigation protection. The project moves beyond simply asking whether an attack is present and instead considers which satellites are suspicious, how much confidence should be placed in the current navigation solution, and how the receiver can respond when that confidence decreases.

The experimental results demonstrate that the approach can substantially reduce positioning errors under controlled step and drift spoofing conditions. The fusion detector achieved perfect recall (1.000) against the step attack and improved recall from 0.967 to 0.989 against the drift attack relative to a temporal-only baseline (Section 8.3). At the navigation level, the reduction of mean drift-induced positioning error from approximately 6289.44 m to approximately 2.94 m demonstrates the practical value of coupling threat assessment with navigation fallback (Section 9.3). This result should be read as a demonstration of the mechanism’s controlled-environment effectiveness rather than a claim of equivalent real-world performance.

At the same time, the weaker detection performance against the evasive attack (fusion recall 0.089) establishes a clear research gap and prevents the system from being presented as universally effective. This balance between demonstrated capability and explicitly identified limitations (Section 12) is central to the project’s value as a practical prototype rather than a finished operational system.

NavIC-SHIELD therefore serves as a foundation for further research into AI-assisted GNSS integrity monitoring, satellite-level threat intelligence, and resilient positioning for denied and degraded environments. The combination of controlled simulation, adversarial attack modelling, spatio-temporal detection, satellite-wise intelligence, navigation confidence assessment, and an interactive deployed dashboard provides a complete experimental platform for studying the relationship between GNSS signal integrity and resilient navigation.

14 Future Work

To advance NavIC-SHIELD toward a validated, field-ready system, the following directions are proposed:

1. **Real-World Data Integration:** Integrate larger real-world GNSS and NavIC datasets and, where possible, physical receiver measurements, allowing the simulated observations and controlled attack models to be compared against real receiver behaviour (addressing the limitation in Section 12.3).
2. **Advanced Temporal Modelling:** Extend the detection system with more advanced temporal models capable of identifying long-horizon and low-amplitude attacks, directly targeting the evasive-attack detection gap identified in Section 8.3 and Section 12.2. Adaptive detection thresholds and online learning could further improve the ability to distinguish environmental degradation from adversarial manipulation.
3. **Multi-Constellation and Multi-Sensor Navigation:** Combine NavIC with other GNSS constellations and integrate inertial measurements to provide additional redundancy when individual satellite measurements become unreliable.
4. **Real-Time Embedded Implementation:** Optimize the detection and navigation pipeline for computationally constrained hardware, and evaluate its latency, memory requirements, and robustness under continuous operation.
5. **Field-Level Testing:** Conduct field-level testing in challenging terrain and degraded navigation environments as a step toward evaluating whether the proposed approach can transition from a controlled research framework into practical navigation-integrity applications.